

Special Topic Area – DARPAVERSE

DARPA-PS-25-05-ST01

Program Competition

BACKGROUND:

The Defense Advanced Research Projects Agency (DARPA) Expedited Research Innovation System (ERIS) is a long-term, open call for new and novel technology solutions. The DARPA ERIS program leverages a "Special Topic Area" process which allows DARPA the opportunity collect submissions against specific problem sets that fall outside of the normal ERIS enduring Topic Areas. For this special topic area (STA), DARPA seeks solutions or capabilities that deliver breakthrough technological advancements for the "DARPAVERSE" program. The DARPAVERSE program aims to create a collaborative environment transforming how warfighters, analysts, and program managers interact with complex modeling systems, developing and deploying models of complex multi-domain problems iteratively to improve assumptions and solution quality with each iteration, approaching a Pareto-frontier of dominating solutions.

This program seeks to achieve a new, iterative capability to develop multi-domain models of concepts and scenarios in less than a day. To enable this capability, DARPAVERSE will establish a comprehensive model repository, along with methods to compose these models into orchestrated workflows using dependent type systems to understand correct composition using natural language interfaces that bridge communication gaps between technical and non-technical users.

**** Additional information****

Virtual Proposers Day video will be posted on the ERIS website.

This Special Notice revision will be posted when the proposer day video live on ERIS.

The "DARPAVERSE" STA is a limited-period announcement to solicit, collect, and assess disruptive solutions to support the technical areas and program goals provided below.

This STA will be open for submissions from 1 October 2025 through 31 December 2025.

The ERIS Master Solicitation can be found here:

<https://sam.gov/opp/fabda3a3d150457d97068977672ec750/view>

WARNING: DO NOT SUBMIT VIA THIS PAGE: This Special Topic Opportunity Page provides the content requirements and instructions for your submission. The actual submission, however, will be made through the ERIS Marketplace located here:

<https://www.darpaconnect.us/eris>. After navigating to the ERIS Marketplace Page, click the "Request ERIS Account" button to register for an ERIS account. Once registered, click the "Access ERIS" button to begin the correct submission workflow.

Problem Statement:

DARPAVERSE: A unified platform enabling accelerated capability development, deployment, and optimal warfighter courses of action:

The Strategic Technology Office (STO) develops technology in systems science to give national security leaders trusted, disruptive capabilities to win in all physical domains – air, space, sea, and land. The end DARPAVERSE capability will seek to deliver a unified platform and immersive environment for rapid concept exploration to enable the warfighter, program manager, and analyst to ask iterative questions and to learn about problem spaces faster, converging on disruptive solutions. STO seeks organizations to support three complementary technical area (TA) efforts to deliver decision superiority to the DoD for capabilities development through rapid, iterative, ideation to discover optimal solutions to difficult problem spaces, shrinking development time from years to weeks:

- **Technical Area 1: Adaptive Software Repository**
- **Technical Area 2: Adaptive Co-Design and Orchestration**
- **Technical Area 3: Adaptive Definition and Visualization**

Organizations may propose to one or more technical areas. Organizations proposing to multiple technical areas must submit a separate video for each TA. This allows for focused evaluation of technical approaches specific to each area. Organizations specializing in one domain are encouraged to apply individually or partner with other organizations. These technologies will be integrated with a government provided system for software defined orchestration over containerized model instances in a standardized API.

Purpose

DARPAVERSE challenges the current operating assumption that multi-domain models complex scenarios and concepts must be derived from stove-piped and monolithic simulation environments that result in models that are not maintainable, not factorable, and not composable with other models of differing provenance. DARPAVERSE plans to disrupt these assumptions by producing a new, open environment that utilizes automated reduced-order model generation to accelerate existing models, derive models from data, and create new representations of those models that can be decomposed and recomposed to solve new problems and represent new assumptions as they arise.

Technical Areas

Technical Area 1: Adaptive Software Repository

TA1 performers will develop advancements in reduced-order modeling techniques similar to Koopman operator theory, dynamic mode decomposition (DMD), and kernel methods to create generalized mechanisms for approximately representing complex, non-linear, dynamical systems. These methods will be used to derive reduced-order models from existing models or data of system dynamics to be represented in the DARPAVERSE in composable representations that have interoperable observables, allowing them not only to

be composed, but also ensuring that the composition of these models is also composable.

Successful submissions will address these problems, enabling the creation of a software repository that will serve as a repository for rapid model development in the DARPAVERSE. Strong TA 1 submission should indicate how they will integrate with this system by automatically generating composable reduced order-models from data provided by previously containerized models, without access to the code.

Technical Area 2: Adaptive Co-Design and Orchestration

TA2 performers will advance research in formal methods and dependent type theory to enable adaptive co-design of composed models. This involves multi-objective optimization where models from the TA1 software repository can be orchestrated into dynamic workflows while maintaining correctness and completeness. Models of different domains, even different versions of the same domain, are not interoperable by default. They may have different assumptions, units, or requirements that prevent their composition. TA2 performers will take common observable spaces generated by TA1 reduced order models and implement a dependent type system that can successfully compose these models or return counterexamples that identify incompatibilities to speed iteration towards a correct composition.

TA2 performers will also research new methods that allow dependent-type systems to be used as constraints that define complex relationships between models, their parameters, units, boundary conditions, requirements, and outputs. Successful submissions will provide a mathematical structure that allows for automatic composition in a structured way, allowing reduced-order models to be composed into multi-domain simulations, while enforcing correctness, completeness, and consistency on the resulting composed systems. The resulting TA2 multi-objective design problem and orchestration system enables rapid development of correct and compatible models that can be formed and modeled into complex systems. Strong TA 2 submissions should indicate how they will create a dependent-type layer capable of checking for the correctness of an orchestration of TA 1 models in the integrated prototype.

Technical Area 3: Adaptive Definition and Visualization

TA3 will address current challenges in usability by providing a natural language interface for generating orchestrated workflows using TA2 technologies, allowing users without a background in coding to provide semi-formal descriptions of desired multi-domain models. TA3 performers will address problems in language ambiguity to automatically transform natural language requirements and specifications of orchestrated workflows and model initial conditions into analyzable formal representations and specifications in an intermediate language that specifies compositions of elements from the TA1 software repository.

While Large Language Models (LLMs) have the potential with tool use to translate specifications from natural language to structured formal representations, they still struggle with the ambiguity and complexity of natural language. Successful TA3 submissions will address issues with incorrect output and translation, providing means for successful error localization to allow users with no coding experience to identify and correct errors in automatic translation, allowing them to iterate quickly in developing multi-domain models and simulations. Strong TA3 submissions should indicate how they will integrate for both model definition, and reporting outcomes and errors of the orchestrated system.

The ERIS rubric can be found as appendix (c) to the master program solicitation.

SUBMISSION INSTRUCTIONS:

The DARPAVERSE program uses a multi-step selection process. The submission process for this Special Topic consists of the following steps:

- **Step I** – Video pitch submission through the ERIS Marketplace Opportunity page.
- **Step II** – Video submissions are assessed in accordance with master program solicitation rubric informed by the specific instructions below.

Submissions will only be accepted via the ERIS Marketplace. After navigating to the ERIS Marketplace Page, click the "Request ERIS Account" button to register for an ERIS account. Once registered, click the "Access ERIS" button to begin the submission workflow.

- Submit your video pitch to the STA: "DARPAVERSE"
- To identify that your video submission is being submitted against the "DARPAVERSE" STA, the title of your submission should start with "DARPAVERSE" and then continue with your company-specific title (e.g., DARPAVERSE: Rapid Experimentation), ending with the Technical Area you are submitting to.
- Submission Title Format: 'DARPAVERSE: [Your Company Title] - TA[X]' (where X = 1, 2, or 3 for the specific technical area being addressed in this video)
- Multi-TA Submissions: Organizations proposing solutions for multiple technical areas must submit separate videos for each TA, using the appropriate TA number in each submission title.

All submissions must comply with Section VII submission requirements of the ERIS Master Program Solicitation (DARPA-PS-25-05), including the 11-point compliance review criteria. The DARPAVERSE-specific requirements below are supplemental to, not replacements for, the standard ERIS requirements. The ERIS submission checklist is available via the ERIS Marketplace website. Each submission must include:

(1) 7-minute pitch video

- Details provided on pp. 3-8, section VII of the program solicitation. Note: Each video must focus on one specific Technical Area. Multi-TA proposers must submit separate videos for each area.

(2) ERIS supplemental slides:

- Details provided on pp. 6-7, section VII of the program solicitation.

(3) Submission form

- Details provided on pp. 3-6, section VII of the program solicitation.

To be considered eligible for a DARPAVERSE award, you must participate in all selection events. Submitted video pitches will be evaluated through the ERIS assessment process. Submitters assessed as "Awardable" through the marketplace may be invited to participate in DARPAVERSE program negotiation event. This event will consist of a presentation of supplemental technical and pricing information, followed by a Q&A session. Following the

program negotiation event, the Government intends to make selections for funding.

If your video is selected into the ERIS marketplace, DARPA will provide additional information in anticipation of the DARPAVERSE program negotiation event. Please consider submitting ERIS video pitches and slides **early** within the open submission period.

Advantages of early ERIS submissions maximize preparation time for the DARPAVERSE program negotiation event.